

Team-11

Project Execution Screenshots:

```
amulya@amulya: ~  
amulya@amulya:~$ nano prol.c  
amulya@amulya:~$ gcc prol.c  
amulya@amulya:~$ ./a.out  
  
=====   
PROCESS TERMINATION COMMAND CENTER  
=====   
Operating System Process Monitoring Project  
Parent Monitor : ACTIVE  
  
=====   
MAIN MENU  
=====   
1. Process Event Simulator  
2. Termination Report Center  
3. Exit  
=====   
Enter choice: |
```

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amulya@amulya:~$ nano prol.c  
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PROCESS TERMINATION COMMAND CENTER  
=====   
Operating System Process Monitoring Project  
Parent Monitor : ACTIVE  
  
=====   
MAIN MENU  
=====   
1. Process Event Simulator  
2. Termination Report Center  
3. Exit  
=====   
Enter choice: 1  
  
=====   
PROCESS EVENT SIMULATOR  
=====   
1. Data Processing Service  
2. File Backup Service  
3. Network Service Failure  
4. Database Service  
5. Security Monitoring Service  
6. Back to Main Menu  
=====   
Select Event: |
```

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4. Database Service
5. Security Monitoring Service
6. Back to Main Menu

Select Event: 1

EVENT 1 : Data Processing Service

PROCESS CREATION

Child PID created : 537
Parent PID : 532

PARENT MONITOR

Waiting for child process termination...
Child process created.
Child PID : 537
Parent PID : 532

TASK EXECUTION

Initializing data processing...
Reading records...
Processing 1000 records...
Data processing completed.

TERMINATION ANALYSIS

Termination Type : NORMAL
Exit Status : 10

WIFEXITED() : TRUE
WEXITSTATUS() : 10

PROCESS CLEANUP

waitpid() : SUCCESS
Process Reaped : YES
Zombie Process : NO

EVENT RESULT

[SUCCESS] Data Processing Service completed normally.
[SUCCESS] Parent collected the termination status.



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5. Security Monitoring Service
6. Back to Main Menu

Select Event: 2

EVENT 2 : File Backup Service

PROCESS CREATION

Child PID created : 540
Parent PID : 532

PARENT MONITOR

Waiting for child process termination...
Child process created.
Child PID : 540
Parent PID : 532

TASK EXECUTION

Initializing backup service...
Scanning files...
Creating backup...
Backup completed successfully.

TERMINATION ANALYSIS

Termination Type : NORMAL
Exit Status : 20

WIFEXITED() : TRUE
WEXITSTATUS() : 20

PROCESS CLEANUP

waitpid() : SUCCESS
Process Reaped : YES
Zombie Process : NO

EVENT RESULT

[SUCCESS] File Backup Service completed normally.
[SUCCESS] Parent collected the termination status.

PROCESS EVENT SIMULATOR

Select Event: 3

=====

EVENT 3 : Network Service Failure

=====

PROCESS CREATION

Child PID created : 545
Parent PID : 532

PARENT MONITOR

Waiting for child process termination...
Child process created.
Child PID : 545
Parent PID : 532

TASK EXECUTION

Starting network service...
Establishing connection...
Processing network requests...

[!] Simulating network service failure...

=====

TERMINATION ANALYSIS

=====

Termination Type : ABNORMAL
Signal : SIGTERM
Signal Number : 15

WIFSIGNALED() : TRUE
WTERMSIG() : 15

PROCESS CLEANUP

waitpid() : SUCCESS
Process Reaped : YES
Zombie Process : NO

EVENT RESULT

[ALERT] Network Service Failure terminated by SIGTERM.
[SUCCESS] Parent collected the termination status.

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PROCESS EVENT SIMULATOR

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5. Security Monitoring Service
6. Back to Main Menu

Select Event: 4

EVENT 4 : Database Service

PROCESS CREATION

Child PID created : 546
Parent PID : 532

PARENT MONITOR

Waiting for child process termination...
Child process created.
Child PID : 546
Parent PID : 532

TASK EXECUTION

Starting database service...
Connecting to database...
Executing database operation...
Updating records...
Database operation completed.

TERMINATION ANALYSIS

Termination Type : NORMAL
Exit Status : 40

WIFEXITED() : TRUE
WEXITSTATUS() : 40

PROCESS CLEANUP

waitpid() : SUCCESS
Process Reaped : YES
Zombie Process : NO

EVENT RESULT

[SUCCESS] Database Service completed normally.
[SUCCESS] Parent collected the termination status.

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- 4. Database Service
- 5. Security Monitoring Service
- 6. Back to Main Menu

Select Event: 5

EVENT 5 : Security Monitoring Service

PROCESS CREATION

Child PID created : 551
Parent PID : 532

PARENT MONITOR

Waiting for child process termination...
Child process created.
Child PID : 551
Parent PID : 532

TASK EXECUTION

Starting security monitor...
Scanning active processes...
Checking system activity...
Security scan completed.

TERMINATION ANALYSIS

Termination Type : NORMAL
Exit Status : 50

WIFEXITED() : TRUE
WEXITSTATUS() : 50

PROCESS CLEANUP

waitpid() : SUCCESS
Process Reaped : YES
Zombie Process : NO

EVENT RESULT

[SUCCESS] Security Monitoring Service completed normally.
[SUCCESS] Parent collected the termination status.

```
=====
                        PROCESS EVENT SIMULATOR
=====
```

- 1. Data Processing Service
- 2. File Backup Service
- 3. Network Service Failure
- 4. Database Service
- 5. Security Monitoring Service
- 6. Back to Main Menu

Select Event: 6

MAIN MENU
1. Process Event Simulator
2. Termination Report Center
3. Exit

Enter choice: |

Select Event: 6

MAIN MENU
1. Process Event Simulator
2. Termination Report Center
3. Exit

Enter choice: 2

```
=====
                        TERMINATION REPORT CENTER
=====
```

- 1. Latest Event Report
- 2. All Executed Events
- 3. Process Statistics
- 4. Back to Main Menu

Enter choice: |

MAIN MENU

1. Process Event Simulator
2. Termination Report Center
3. Exit

Enter choice: 2

TERMINATION REPORT CENTER

1. Latest Event Report
2. All Executed Events
3. Process Statistics
4. Back to Main Menu

Enter choice: 1

LATEST EVENT REPORT

Event	: 5
Service	: Security Monitoring Service
Child PID	: 551
Parent PID	: 532
Termination	: NORMAL
Exit Status	: 50
Process Reaped	: YES
Zombie Process	: NO


```
=====
                        TERMINATION REPORT CENTER
=====
```

- ```
1. Latest Event Report
2. All Executed Events
3. Process Statistics
4. Back to Main Menu
=====
```

```
Enter choice: 2
=====
```

```
=====
 EVENT HISTORY
=====
```

```
Event 1 : Data Processing Service
```

```
PID : 537
Result : NORMAL
Exit Code : 10
Reaped : YES
```

```
Event 2 : File Backup Service
```

```
PID : 540
Result : NORMAL
Exit Code : 20
Reaped : YES
```

```
Event 3 : Network Service Failure
```

```
PID : 545
Result : SIGNAL
Signal : SIGTERM (15)
Reaped : YES
```

```
Event 4 : Database Service
```

```
PID : 546
Result : NORMAL
Exit Code : 40
Reaped : YES
```

```
Event 5 : Security Monitoring Service
```

```
PID : 551
Result : NORMAL
Exit Code : 50
Reaped : YES
=====
```

```

=====
=====
 TERMINATION REPORT CENTER
=====
1. Latest Event Report
2. All Executed Events
3. Process Statistics
4. Back to Main Menu
=====
Enter choice: 3
=====
 PROCESS STATISTICS
=====
Events Executed : 5
Normal Terminations : 4
Signal Terminations : 1
Processes Reaped : 5
Zombie Processes : 0

Monitoring Status : ACTIVE
=====

```

```

=====
=====
 TERMINATION REPORT CENTER
=====
1. Latest Event Report
2. All Executed Events
3. Process Statistics
4. Back to Main Menu
=====
Enter choice: 4
=====
MAIN MENU
=====
1. Process Event Simulator
2. Termination Report Center
3. Exit
=====
Enter choice: |

```

```

MAIN MENU
=====
1. Process Event Simulator
2. Termination Report Center
3. Exit
=====
Enter choice: 3

Shutting down Process Termination Command Center...
Program terminated successfully.
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```